DO NOW — RECALL

Interrupts and service routines

Name: _____ Class: _____ Date: _____

1. Define fetch-decode-execute.

2. What causes a process to enter Waiting?

3. What does an OS scheduler choose?

4. Simplify (A∧B)∨(A∧¬B), showing factoring.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Read, interpret, execute an instruction.

2. Model answer

Unavailable I/O or required event.

3. Model answer

Which Ready process receives CPU time.

4. Model answer

A∧(B∨¬B)=A∧1=A.

Vocabulary — know these terms

interrupt — CPU attention request

ISR — Interrupt Service Routine

context — State for safe resume

polling — Repeatedly checking event state

NEW LEARNING

Read, annotate and explain

Understand

An interrupt signals an event such as device completion to the processor.

Trace

The CPU saves sufficient execution context before servicing the event.

Explain

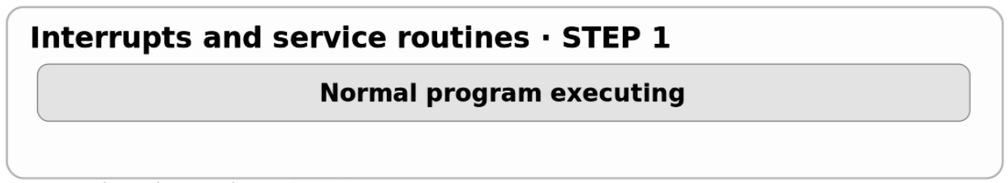
An ISR handles the interrupt, acknowledges it, then restores normal execution state.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

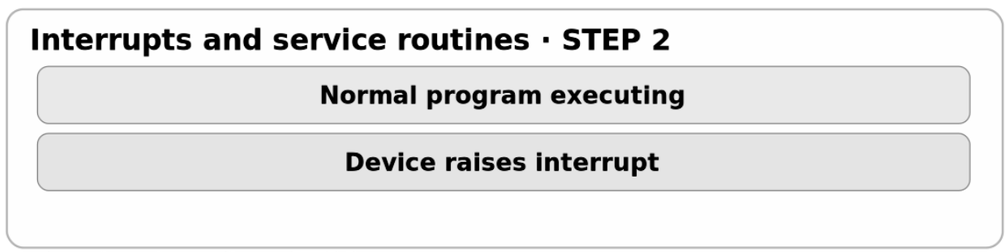
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



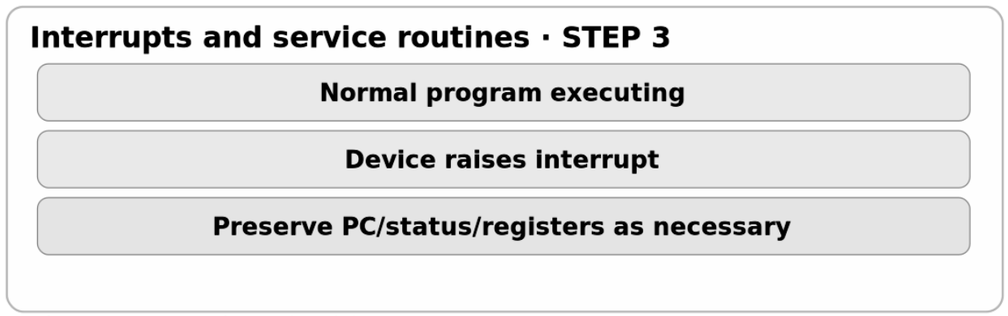
Interrupts and service routines · STEP 1

New Learning



Interrupts and service routines · STEP 2

New Learning



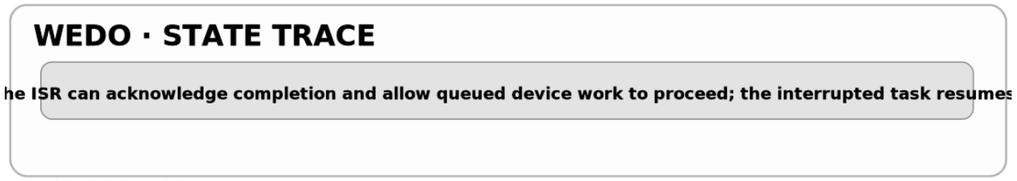
Interrupts and service routines · STEP 3

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A printer signals I/O completion. Describe ISR and scheduler actions.

The ISR can acknowledge completion and allow queued device work to proceed; the interrupted task resumes.



WEDO · STATE TRACE

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

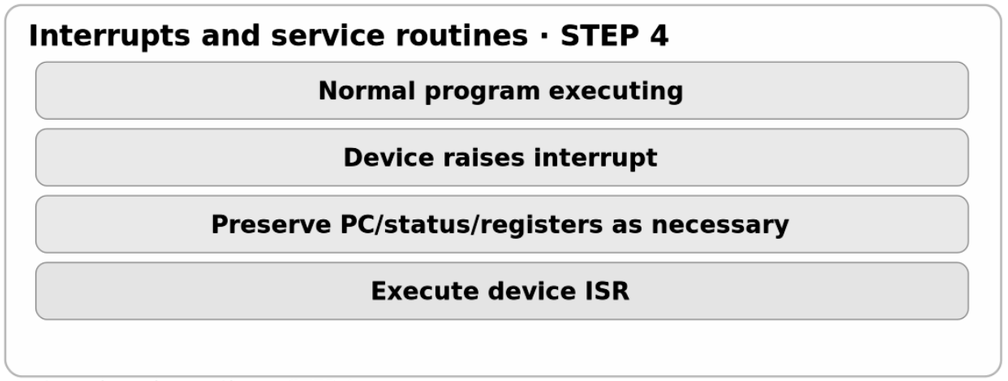
Guided 1 — CHECK

Predict first
The ISR can acknowledge completion and allow queued device work to proceed; the interrupted task resumes.

New Learning · original lesson diagrams

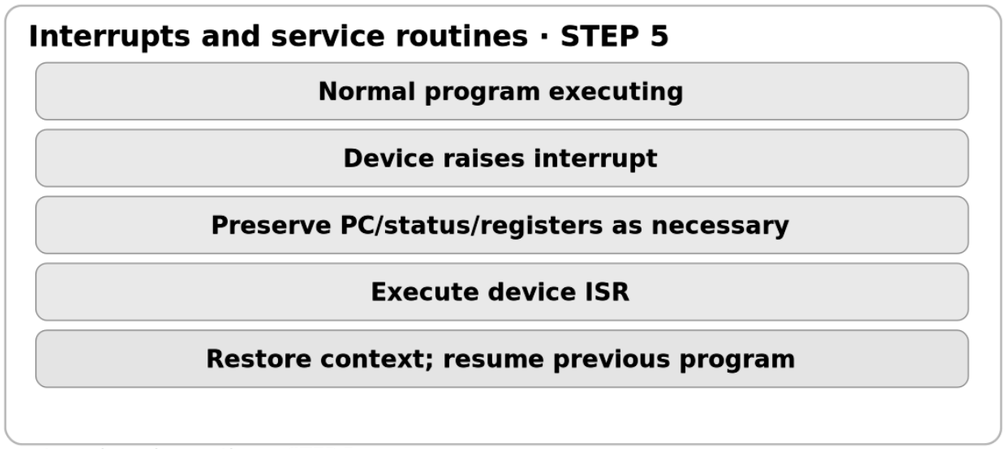
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Interrupts and service routines · STEP 4

New Learning



Interrupts and service routines · STEP 5

I DO — WORKED EXAMPLE

Follow the worked method

Keyboard triggers an interrupt during a text editor. Trace the stages.

Predict first

The ISR processes keyboard input before restoring the prior program state.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

An interrupt handler can overwrite original registers without preserving any state.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare polling and interrupt-driven I/O for infrequent device input. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Keyboard triggers an interrupt during a text editor. Trace the stages.

The ISR processes keyboard input before restoring the prior program state.



IDO · STATE TRACE

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A printer signals I/O completion. Describe ISR and scheduler actions.
Source code / input:
Printer complete → interrupt → ISR acknowledges → spool next job

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Define interrupt, ISR and one source of interrupts. [4]

Question 2

Why does CPU context need to be saved/restored around a device ISR? [6]